

- 1 **Exercise:** A sample of size 50 is drawn ^{iid} from a population with unknown
 2 mean and variance. It holds that $\bar{x} = 14.6$ and $s = 1.3$. Estimate the
 3 population mean. State both the standard error for the estimator and a
 4 90% confidence interval for μ .

5 $\bar{x} = 14.6$ is our best estimate of μ

6 "Our best estimate of μ is 14.6. It was found
 7 using an estimator known to be unbiased."

8 know $V(\bar{x}) = \sigma^2/n$, so approx. SE as $s/\sqrt{n} = 1.3/\sqrt{50}$

9 "The SE of this estimator is $1.3/\sqrt{50}$. So, I
 10 should expect errors in this estimate on
 11 the order of $1.3/\sqrt{50}$."

12 Since n large enough, can appeal to CLT
 to construct a CI (first case listed above).

$$\text{90\% CI is } \bar{x} \pm \underset{\substack{\uparrow \\ \text{sp. norm. ppf}(.95, 0, 1)}}{z_{.05}} \left(\frac{s}{\sqrt{n}} \right), \quad 14.6 \pm (1.645) \left(\frac{1.3}{\sqrt{50}} \right)$$

"I am 90% confident that the true value of μ
 is between"

1 Part 4: Constructing Estimators

2 Summary

3 Recall that typically a probability model is **incompletely specified**. I.e.,
4 a particular model is assumed, but there are still unknown components,
5 often real-valued **parameters**. We use available data (a **sample**) to estimate
6 these parameters.

7 In this Part we describe methods for constructing estimators for unknown
8 parameters.

1 Terminology Review

2 The **parameters** are the constants that distinguish probability models in the
3 same “family.” Examples of “families” include “normal,” “exponential,”
4 and so forth.

5 A **statistic** is a function of the RVs being used to model the observable data.
6 Examples include the sample mean, the sample variance, etc. A statistic is
7 a **random variable**.

8 An **estimator** is a statistic which is used to estimate an unknown parame-
9 ter. It is also a **random variable**.

10 The “hat notation” is standard for denoting an estimator: $\hat{\beta}$ is an estimator
11 for β , for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its
12 estimator.

- 1 An estimator $\hat{\theta}$ is said to be **an unbiased estimator for θ** if $E(\hat{\theta}) = \theta$.
- 2 The variability in an estimator (its precision) is naturally quantified by its
- 3 variance, $V(\hat{\theta})$.
- 4 The **standard error (SE) of $\hat{\theta}$** is the square root of its variance. The SE is of-
- 5 ten reported along with an estimate in order to give a sense of the amount
- 6 of error one should anticipate in that estimate.

1 **The Method of Moments**

- 2 Simple cases such as estimating the population mean and estimating the
- 3 population proportion can be misleading: In those cases it is “obvious”
- 4 what the natural estimator is, since one can use the sample mean or sample
- 5 proportion as the estimator, respectively.
- 6 In other cases, the choice of estimator is not so clear.
- 7 We first consider a simple approach to constructing estimators, the **method**
- 8 **of moments**.

1 First, recall that for any positive integer k , the quantity $E(X^k)$ is called the
 2 k^{th} moment for the distribution of X . We might call this the “ k^{th} population
 3 moment” to be more clear.

4 Imagine that I have random variables $X_1, X_2, \dots, X_n$ that model the ob-
 5 servable sample. Then the k^{th} sample moment is

6
$$M_k = \sum_{i=1}^n X_i^k / n$$

7 Note that M_k is our best estimator for $E(X^k)$, for instance $E(M_k) = E(X^k)$

1 Now, suppose that $X_1, X_2, \dots, X_n$ are i.i.d. each with density $f_X(x; \theta)$.

2 Here, $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ is a vector of parameters.

3 Set up a system of p equations, setting the first p sample moments equal to
 4 the first p population moments:

5
$$M_1 = E(X)$$

6
$$M_2 = E(X^2)$$

7
$$\vdots$$

8
$$M_p = E(X^p)$$

9
$$\underbrace{M_1, \dots, M_p}_{\text{functions of } X_1, X_2, \dots, X_n} = \underbrace{E(X), \dots, E(X^p)}_{\text{functions of } \theta}$$

10 We note that the sample moments are all quantities we can determine from
 11 the available sample. We also note that the population moments are func-
 12 tions of the unknown parameters.

1 We can now solve this system of equations for $\theta_1, \theta_2, \dots, \theta_p$:

$$\begin{aligned} 2 \quad \theta_1 &= g_1(X_1, X_2, \dots, X_n) \\ 3 \quad \theta_2 &= g_2(X_1, X_2, \dots, X_n) \\ 4 \quad \vdots &\quad \vdots \\ 5 \quad \theta_p &= g_p(X_1, X_2, \dots, X_n) \end{aligned}$$

7 These equations are built on the (incorrect) belief that $M_i = E(X^i)$.

8 In fact, $M_i \approx E(X^i)$, so

$$9 \quad \theta_i \approx g_i(X_1, X_2, \dots, X_n)$$

10 and we construct our estimator as

$$11 \quad \underline{\hat{\theta}_{i,\text{MOM}}} = g_i(X_1, X_2, \dots, X_n)$$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. from the Exponential(λ)
 2 distribution. (Recall that this means that $E(X_i) = 1/\lambda$.) What is the method
 3 of moments estimator for λ ?

$$\begin{aligned} 4 \quad & p=1, \text{ so only need one equation} \\ 5 \quad & M_1 = \sum X_i / n = \bar{X} \quad \text{first sample moment} \\ 6 \quad & E(X) = 1/\lambda \quad \text{first pop. moment} \\ 7 \quad & \text{System of equations: } M_1 = E(X) \\ 8 \quad & \bar{X} = 1/\lambda \\ 9 \quad & \text{Solve for } \lambda: \lambda = 1/\bar{X}, \text{ so } \hat{\lambda}_{\text{mom}} = 1/\bar{X} \end{aligned}$$

- 1 **Exercise:** What is the result if, instead of the first moment, the second mo-
 2 ment is used to construct the estimator for λ in the previous example?

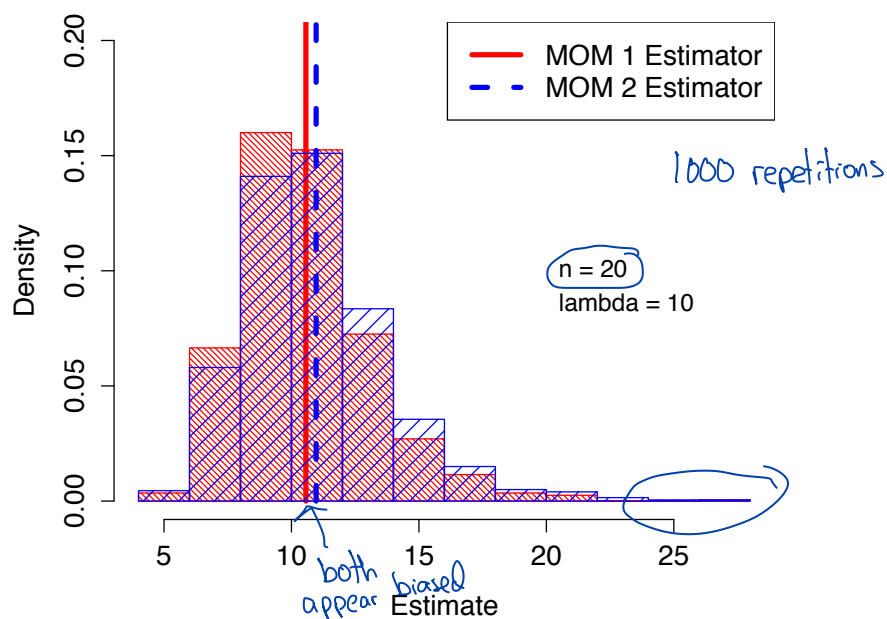
$$E(X^2) = V(X) + E(X)^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2}$$

$$M_2 = \frac{\sum X_i^2}{n}$$

$$\text{System of equations: } \frac{\sum X_i^2}{n} = \frac{2}{\lambda^2}$$

$$\text{Solve for } \lambda, \text{ get } \hat{\lambda}_{\text{mom},2} = \sqrt{\frac{2n}{\sum X_i^2}}$$

Comparison of MOM Estimators



- 1 **Exercise:** Is the method of moments estimator found above (using the first
 2 moment) an unbiased estimator for λ ? Can you adjust it to make it an
 3 unbiased estimator?

4 We know $\bar{X} \sim \text{Gamma}(n, n\lambda)$

5 So $1/\bar{X} \sim \text{Inverse Gamma}(n, n\lambda)$

6 If $Y \sim \text{InvGamma}(\alpha, \beta)$, then $E(Y) = \beta/(\alpha-1)$, $\alpha > 1$

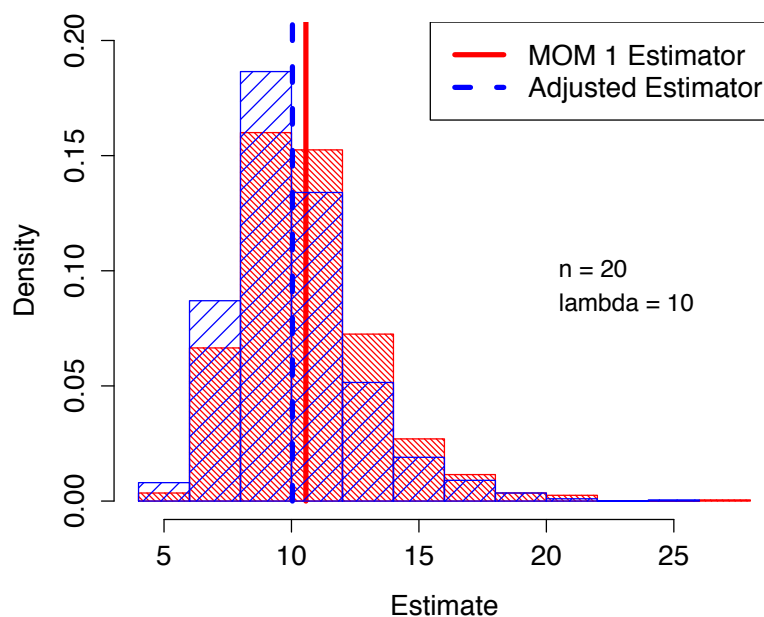
7 So, $E(\hat{\lambda}_{\text{mom},1}) = E(1/\bar{X}) = \frac{n\lambda}{n-1} = \left(\frac{n}{n-1}\right)\lambda \neq \lambda$

9 But, $\hat{\lambda}_3 = \left(\frac{n-1}{n}\right)\hat{\lambda}_{\text{mom},1}$ is unbiased for λ

11 Also, $V(\hat{\lambda}_3) = \left(\frac{n-1}{n}\right)^2 V(\hat{\lambda}_{\text{mom},1}) < V(\hat{\lambda}_{\text{mom},1})$

12 So, $\text{MSE}(\hat{\lambda}_3) < \text{MSE}(\hat{\lambda}_{\text{mom},1})$

Comparison of MOM Estimator and Adjusted MOM



- 1 The comparisons made above between the three estimators are based **en-**
 2 **tirely on simulations**. There is **no** observed data involved at all. This is
 3 the **frequentist approach** to assessing estimators: **how does the estimator**
 4 **perform in repeated use?**
- 5 As seen above, simulations are a useful tool for cases where the mean and
 6 variance derivations are not available.

- 1 **Exercise:** Interpret the output below. What is the approximate MSE of each
 2 of the three estimators?

```
> print(c(mean(mom1), mean(mom2), mean(adju)))
[1] 10.56553 10.96759 10.03725
```

```
> print(c(var(mom1), var(mom2), var(adju)))
[1] 6.574420 8.224773 5.933414
```

3 $S_0, E(\hat{\tau}_{mom_2}) \approx 10.97, V(\hat{\tau}_{mom_2}) \approx 8.22$
 4 $MSE(\hat{\tau}_{mom_2}) \approx (10.97 - 10)^2 + 8.22$

5 $MSE(\hat{\tau}_{adj}) \approx 5.93$
 6
 7
 8

1 Maximum Likelihood Estimation

2 **Maximum likelihood estimation (MLE)** is a powerful, general approach to
3 constructing estimators.

4 We will begin the explanations with simple, one-dimensional examples,
5 but the concepts extend to complex, high-dimensional models in a natural
6 manner.

7 Importantly, there is also a general way of calculating standard errors of
8 MLEs, and, under a wide range of conditions, the MLE is approximately
9 normally distributed. This makes inference with MLEs straightforward.

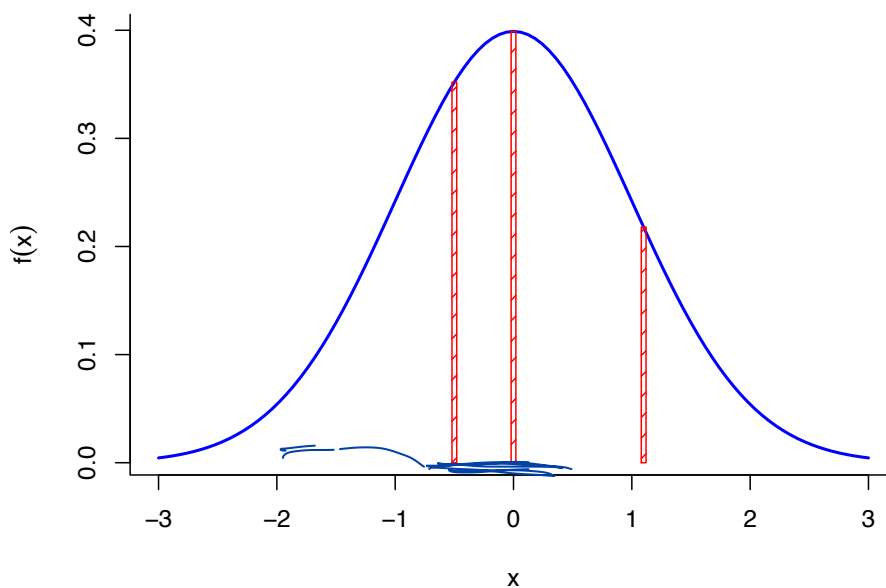
1 Basic Setup

2 Recall that if X is a continuous random variable with density f_X , that does
3 **not** mean that $P(X = a) = f_X(a)$, i.e., the density does **not** return proba-
4 bilities. Indeed, $P(X = a) = 0$ for all a .

5 **It is true** that the density is higher in regions where there is the most
6 probability: For fixed, small $\epsilon > 0$, if we want to find a to maximize
7 $P(a - \epsilon \leq X \leq a + \epsilon)$, then we would look for the a that maximizes $f_X(a)$.

8 See the figure on the following slide. The standard normal density is de-
9 picted. The most likely values are close to zero, where the density is maxi-
10 mized.





1

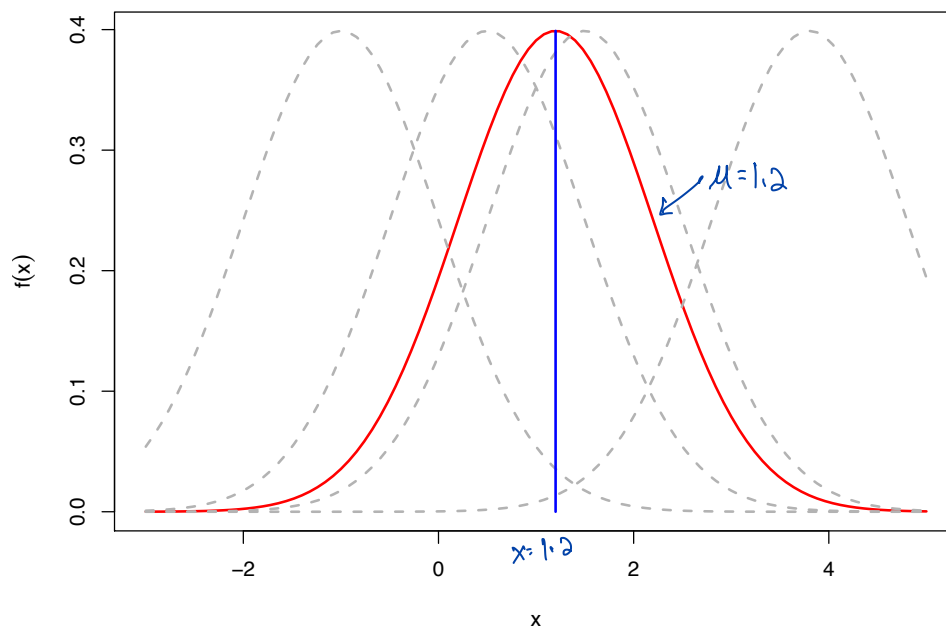
x

1 As previously stated, we use probability distributions such as the normal
2 to “model” the real processes that generate data. So, if we assume that
3 our observable data is drawn from the standard normal distribution, the
4 density above specifies where we are more or less likely to see our data
5 value.

6 We can turn this idea around: Suppose that we observe data $x = 1.2$,
7 and we are willing to assume that it came from a normal distribution with
8 mean μ and variance one. What value of μ makes the observation $x = 1.2$
9 the most likely?

10 See the figure on the following slide. Different choices of μ are displayed.
11 It’s not difficult to conclude that when $\mu = 1.2$, the **likelihood** is maxi-
12 mized.

13



1

x

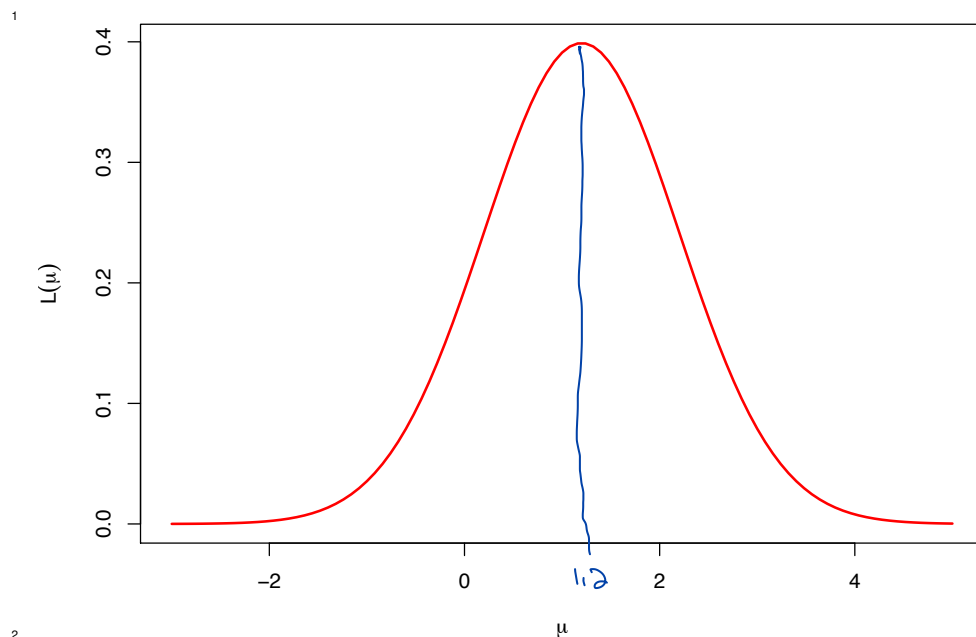
1 The **likelihood function** is constructed by calculating the distribution for
 2 the data, and then **holding the data constant** and allowing the parameter(s) to vary.

4 In the simple example above, we assume that we observe a single data
 5 point x that was generated from a normal distribution with mean μ and
 6 variance one. Hence, the distribution is

$$7 \quad f(x; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2}\right)$$

8 If we fix the data $x = 1.2$, and think of this as a function of μ , then we arrive
 9 at our likelihood function:

$$10 \quad \underline{L(\mu)} = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\overset{x}{(1.2 - \mu)^2}}{2}\right)$$



- 1 The **maximum likelihood estimator** for the parameter is the choice of the
2 parameter which maximizes the likelihood function.
- 3 Heuristically, we ask: Among all possible choices for the parameter, which
4 one makes our observed data the most likely?
- 5 It is worth stressing that the likelihood function is **not** itself a probability
6 distribution. It is **not** specifying the relative “probabilities” for different
7 values of the parameter.
- 8 Remember that in the frequentist approach, the unknown parameter is a
9 constant, and not random.

- 1 **Exercise:** An urn contains black and white balls. All that we know is that
 2 the balls have a color ratio of 1 to 3. (We do not know if there are more
 3 white or more black balls.) We draw three balls **with replacement** from
 4 the urn. Let X equal the number of black draws. Derive the likelihood
 5 function and the MLE as a function of X .

6 $\Theta = P(\text{single draw is black})$, Θ is either $\frac{1}{4}$ or $\frac{3}{4}$
 7 $X \sim \text{binomial}(3, \Theta)$
 8

	$X=0$	$X=1$	$X=2$	$X=3$
$\Theta = \frac{1}{4}$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$
$\Theta = \frac{3}{4}$	$\frac{1}{64}$	$\frac{9}{64}$	$\frac{27}{64}$	$\frac{27}{64}$

 9 $\hat{\Theta}_{MLE} = \begin{cases} \frac{1}{4}, & X=0,1 \\ \frac{3}{4}, & X=2,3 \end{cases}$
 10
 11
 12

- 1 **Exercise:** Now suppose that θ equals the proportion of balls in the urn
 2 which are black. We know that $0 < \theta < 1$. We draw ten times with replace-
 3 ment, and count seven black balls and three white balls. Derive the MLE
 4 for θ .

5 $X = \# \text{ of black draws. Observed } x=7$
 6 $f_X(x) = \binom{10}{x} \theta^x (1-\theta)^{10-x}$, $x=0,1,\dots,10$
 7 The likelihood function when $x=7$ is
 8 $L(\theta) = \binom{10}{7} \theta^7 (1-\theta)^3$
 9 $\frac{\partial L(\theta)}{\partial \theta} = \binom{10}{7} \left[7\theta^6 (1-\theta)^3 + 3\theta^7 (1-\theta)^2 (-1) \right]$
 10
 11 $\frac{\partial L(\theta)}{\partial \theta} = 0$, solve for θ , find $\hat{\theta}_{MLE} = 0.7$ (after checking second deriv.) ← the sample proportion

1 The preceding examples involved a single, observable random variable. In
2 most cases, we have a sequence of random variables in the model.

3 A common case is where $X_1, X_2, \dots, X_n$ are i.i.d. with density $f_X(x; \theta)$,
4 where θ is the parameter to be estimated. In this case, the joint distribution
5 of the random variables is

$$\prod_{i=1}^n f_{X_i}(x_i; \theta)$$

7 and the likelihood function is

$$L(\theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta)$$

1 Finding the MLE Analytically

2 In some situations the MLE can be derived analytically, usually via calcu-
3 lus, but sometimes via direct inspection of the likelihood.

4 One “trick” that is often very helpful: The likelihood function is positive
5 at its maximum. Hence, the likelihood is maximized at the same θ as does
6 the logarithm of the likelihood.

7 Hence, we often work with the **log likelihood** instead of the likelihood.

8 This is often denoted $\ell(\theta) = \log L(\theta)$.

9 In the case where $X_1, X_2, \dots, X_n$ are i.i.d. $f_X(x; \theta)$, then

$$\ell(\theta) = \log \prod_{i=1}^n f_{X_i}(x_i; \theta) = \sum_{i=1}^n \log f_{X_i}(x_i; \theta)$$